3. Pokémon TCG Database Modeling Exercise

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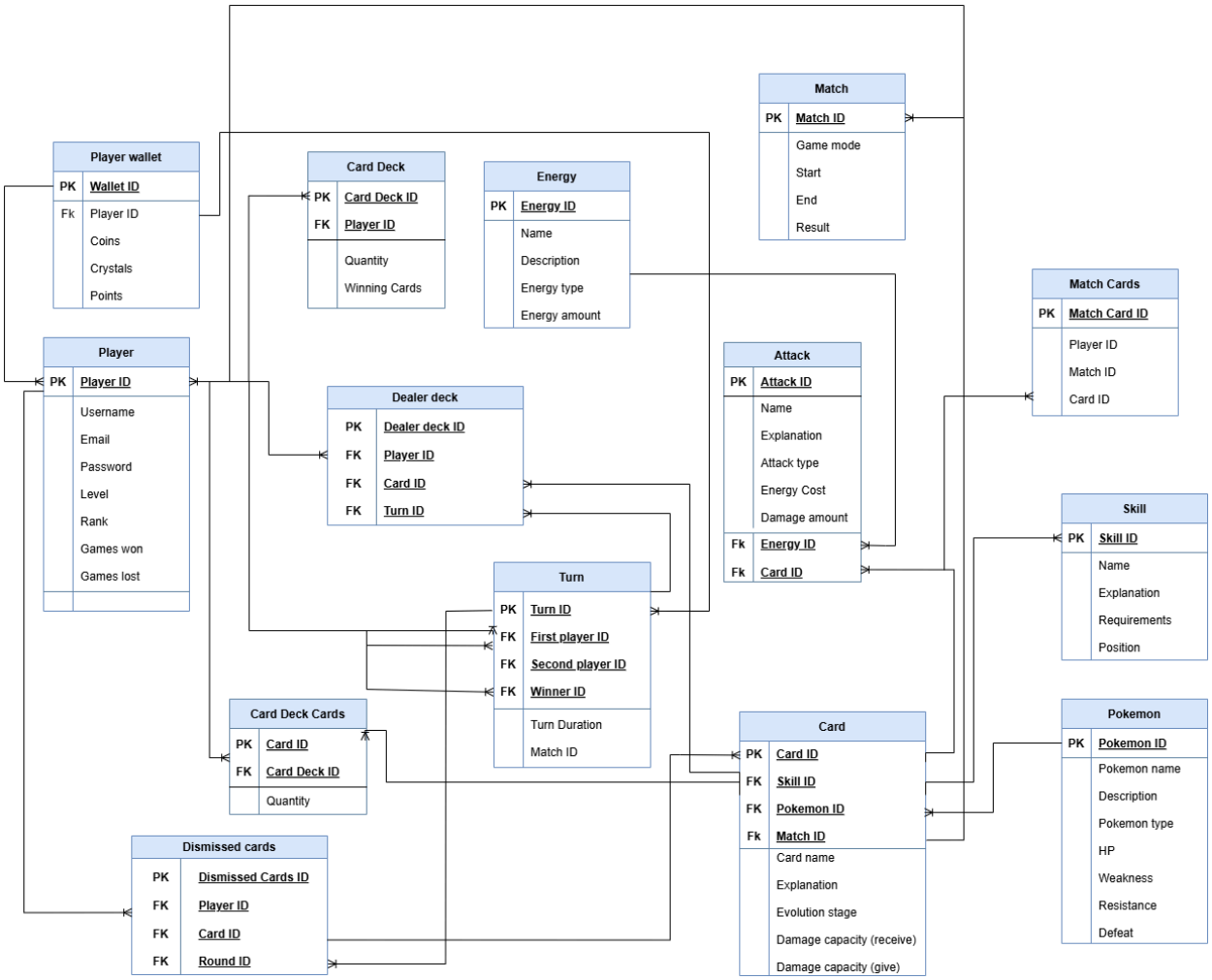
Software Construction and Decision Making

Group 501



Justification:

This is a possible design for a Pokemon TCG game, the main purpose for this diagram was to describe the process the player develops in a basic Pokemon TCG. It was intended to balance the relations between entities and relations. The diagram is meant to be divided into four sections meant to describe the process of the game as clearly as possible; the player system, the deck and cards, the pokemon and skills aspect and finally the gameplay system

Entities:

1. **Player**: The player is the main entity that does not only store the personal information of the user but at the same time it is in charge of tracking the process itself; including the level, ranking and the results themselves.
2. **Cards**: The card includes the elements that will allow the player to to develop throughout the game in the sense of characteristics and uses against other cards.
3. **Matches**: The matches will consider the full game process; it includes various turns and considers the time it starts and ends and the final result.
4. **Turns**: The different rounds throughout the match, consider the different players, the result and the time spent during the game. One match will have various turns and will count both of the players
5. **Card deck**: It is the different cards that each player possesses. One player may have different decks yet their importance relies in identifying both the amount of cards and the stats of the cards and their possibility of winning
6. **Player wallet**: The player's wallet will refer to certain values related to the player and their results. It is important to acknowledge that for each player a single wallet shall exist.
7. **Card deck cards**: This will connect the cards to the decks and the amount of each card on the deck. The main premise of this involves the idea that a single deck will have many cards and one card may exist in many decks.
8. **Pokemon**: It is a requirement to identify the type of pokemon in order to know the cards strengths and weaknesses, since the type will make the result vary.
9. **Skill**: Registering the skill will be needed for registering the abilities that can apply to the different cards and how they should be applied in order to actually achieve good results in battle
10. **Attack**: One card will be able to develop various attacks, having to register the different attacks and the energy they cost, it is very important to keep track of the attacks and their cost in order to determine

11. **Energy**: Identify the type of attacks and how it will be used during attacks. The energy will depend on the amount and type of energy.
12. **Dealer deck**: The concept of the dealer deck will track the card that is dealt to which player during each round. It will help with the card distribution and avoid overloading the Match cards; separating between ownership and the distribution
13. **Dismissed cards**: We should consider the discarded cards, taking in account who discarded it, which card and when in order to keep track of the game. It refers to the process of dismissing rather than the state

Relevant relationships:

1. Player → Player Wallet (1:1): we can separate the game 'economics' from the player itself. It will avoid cluttering the table.
2. Player → Card Deck (1:N): the player is allowed to have various card decks.
3. Card Deck ↔ Card (M:N): a card deck will be able to contain many cards and a single card will be able to appear in different decks.
4. Pokemon → Card (1:N): a pokemon may have various cards and different evolution stages. It avoids duplicating pokemon data across the cards.
5. Skill → Card (1:N): a skill may be available in different cards.
6. Card → Attack (1:N): each card is allowed to have different types of attacks.
7. Energy → Attack (1:N): a single type of energy can be present in multiple attacks. Each attack will consume a specific type and amount of energy.
8. Card → Dismissed cards (1:N): identifying the dismissed card, and ensure that is a valid card
9. Player → Dismissed card (1:N): a player may discard multiple cards, it will track who discarded the card after owning it
10. Turn → Dismissed cards (1:N): in a turn there may be various dismissed cards

11. Match → Dismissed cards (1:N): a card that was dismissed will be part of a match, it should not mix between matches
12. Match → Turn (1:N): a match will have various turns for each player.
13. Match → Match Cards (1:N): a match will involve many cards with different characteristics
14. Card → Match Cards (1:N): a card may appear in different matches
15. Player → Match Cards (1:N): in order to identify who used the cards in the match and keep track of it, avoiding duplication
16. Players (1 and 2) → Turn (1:N): a player may play in different turns
17. Player → Dealer Deck (1:N): a player may receive multiple cards and will ensure the cards to be dealt their cards differently throughout their turn.
18. Card → Dealer Deck (1:N): a card may be in possession of different dealer decks in a turn.
19. Turn → Dealer Deck (1:N): a game's turn can have various dealer decks.

Domain and Constraints:

	Domains	Constraints
Player	Player ID: Integer Username: Varchar (20-30) Email: Varchar (255) Password: Varchar (255) Level: INT Rank: Varchar	Player ID: Primary Key Email: [Unique, NOT NULL] Username: [Unique, NOT NULL]
Player Wallet	Coins, Points and Crystals: Integers	Player ID: Primary Key *There cannot be negative currency
Card	Card ID: INT Name: Varchar (40) Damage: INT	* Name cannot be null
Pokemon	HP: INT Type: Varchar	* Must match the energy type * No duplicate rows * HP cannot be 0

Attack	Damage: INT Energy: INT	* Attacks must correspond to cards
Card Deck	Quantity: INT	* Each deck should have a specific amount of cards
Card Deck Cards	Quantity: INT	Card Deck: Primary Key Card ID: Primary Key * Quantity must be between 1 and 4 * There should not be duplicated rows
Match	Start and End: Datetime	* End must be higher than start stime
Turn	Turn Duration: INT	* First player and second player must be different * Turn must belong to a correct match * Someone must win

Referential:

Primary keys require to exist with their respective entity, and for the foreign keys, they depend on the parent and child connections between each other. In the sense that a child cannot exist without a parent and while there is a parent we cannot erase their link. For example, Match Cards cannot exist unless Match ID exists in the same table